
Table of Contents

.....	1
=====	1
=====	1
Process each file	2
Load file	2
Magnitude	2
Peak / RMS feature	3
Smooth signal	3
Adaptive baseline	3
Event detector settings	3
Find events	3
Timing analysis	3
Accept roughly 120 Hz repeating behavior	4
Final decision	4
Print results	4
Overall validation	5
Comparison plot	5

```
clear;
clc;
close all;
```

```
=====
=====
```

Ali Detector v06 Validate the detector using all three known Talley signals

```
=====
=====
```

```
fileNames = {
    'WN_ONLY_001.mat'
    'SPARK_ONLY_001.mat'
    'SPARK_NOISE_EQUAL_RMS_001.mat'
};
```

```
fileLabels = {
    'White Noise Only'
    'Spark Only'
    'Spark + Noise'
};
```

```
% Ground truth
% 0 = no spark expected
% 1 = spark expected
```

```

expectedResult = [0 1 1];

numFiles = length(fileNames);

detectedResult = zeros(1,numFiles);
peakRMSresults = zeros(1,numFiles);
eventCounts = zeros(1,numFiles);
repetitionRates = zeros(1,numFiles);

fprintf('=====\\n');
fprintf('      ALI DETECTOR v06 VALIDATION TEST\\n');
fprintf('=====\\n\\n');

```

```

=====
      ALI DETECTOR v06 VALIDATION TEST
=====

```

Process each file

```

for n = 1:numFiles

    fprintf('\\n-----\\n');
    fprintf('%s\\n', fileLabels{n});
    fprintf('-----\\n');

```

```

-----
White Noise Only
-----

```

```

-----
Spark Only
-----

```

```

-----
Spark + Noise
-----

```

Load file

```

data = load(fileNames{n});

iq = data.eventIQ;
fs = data.fs;

```

Magnitude

```

mag = abs(iq);

```

Peak / RMS feature

```
rmsLevel = rms(mag);  
peakLevel = max(mag);  
peakRMS = peakLevel / rmsLevel;  
  
peakRMSresults(n) = peakRMS;
```

Smooth signal

```
windowLength = 200;  
smoothMag = movmean(mag, windowLength);
```

Adaptive baseline

```
baseline = median(smoothMag);  
threshold = baseline * 1.5;
```

Event detector settings

```
minPeakDistance = round(0.006 * fs);  
minProminence = 0.006;
```

Find events

```
[~, peakLocations] = findpeaks( ...  
    smoothMag, ...  
    'MinPeakHeight', threshold, ...  
    'MinPeakProminence', minProminence, ...  
    'MinPeakDistance', minPeakDistance);  
  
eventCount = length(peakLocations);  
eventCounts(n) = eventCount;
```

Warning: Invalid MinPeakHeight. There are no data points greater than MinPeakHeight.

Timing analysis

```
repetitionRate = 0;  
repeatingPattern = false;  
  
if eventCount > 1  
    peakTimes = peakLocations / fs;  
    spacing = diff(peakTimes);  
    medianSpacing = median(spacing);
```

```
repetitionRate = 1 / medianSpacing;  
  
repetitionRates(n) = repetitionRate;
```

Accept roughly 120 Hz repeating behavior

```
if medianSpacing > 0.007 && medianSpacing < 0.010  
    repeatingPattern = true;  
end  
  
end
```

Final decision

Require BOTH: 1. impulsive Peak/RMS behavior 2. repeating timing pattern

```
peakRMSThreshold = 8;  
  
if peakRMS > peakRMSThreshold && repeatingPattern  
    detectedResult(n) = 1;  
else  
    detectedResult(n) = 0;  
end
```

Print results

```
fprintf('Peak/RMS: %.2f\n', peakRMS);  
fprintf('Events detected: %d\n', eventCount);  
  
if repetitionRate > 0  
    fprintf('Repetition rate: %.2f Hz\n', repetitionRate);  
else  
    fprintf('Repetition rate: N/A\n');  
end  
  
if detectedResult(n) == 1  
    fprintf('Detector result: CANDIDATE SPARK\n');  
else  
    fprintf('Detector result: NO SPARK\n');  
end  
  
if expectedResult(n) == detectedResult(n)  
    fprintf('Validation: PASS\n');  
else  
    fprintf('Validation: FAIL\n');  
end
```

```
Peak/RMS: 4.04  
Events detected: 0  
Repetition rate: N/A  
Detector result: NO SPARK  
Validation: PASS
```

```
Peak/RMS: 29.83
Events detected: 120
Repetition rate: 120.00 Hz
Detector result: CANDIDATE SPARK
Validation: PASS
```

```
Peak/RMS: 22.70
Events detected: 120
Repetition rate: 119.96 Hz
Detector result: CANDIDATE SPARK
Validation: PASS
```

```
end
```

Overall validation

```
correct = sum(expectedResult == detectedResult);
accuracy = 100 * correct / numFiles;

fprintf('\n\n=====\\n');
fprintf('          FINAL VALIDATION\\n');
fprintf('=====\\n');

fprintf('Correct tests: %d / %d\\n', correct, numFiles);
fprintf('Validation accuracy: %.1f %%\\n\\n', accuracy);

for n = 1:numFiles

    fprintf('%-20s Expected: %d   Detected: %d\\n', ...
        fileLabels{n}, ...
        expectedResult(n), ...
        detectedResult(n));

end
```

```
=====
          FINAL VALIDATION
=====
Correct tests: 3 / 3
Validation accuracy: 100.0 %

White Noise Only      Expected: 0   Detected: 0
Spark Only            Expected: 1   Detected: 1
Spark + Noise         Expected: 1   Detected: 1
```

Comparison plot

```
figure;

bar(peakRMSresults);
```

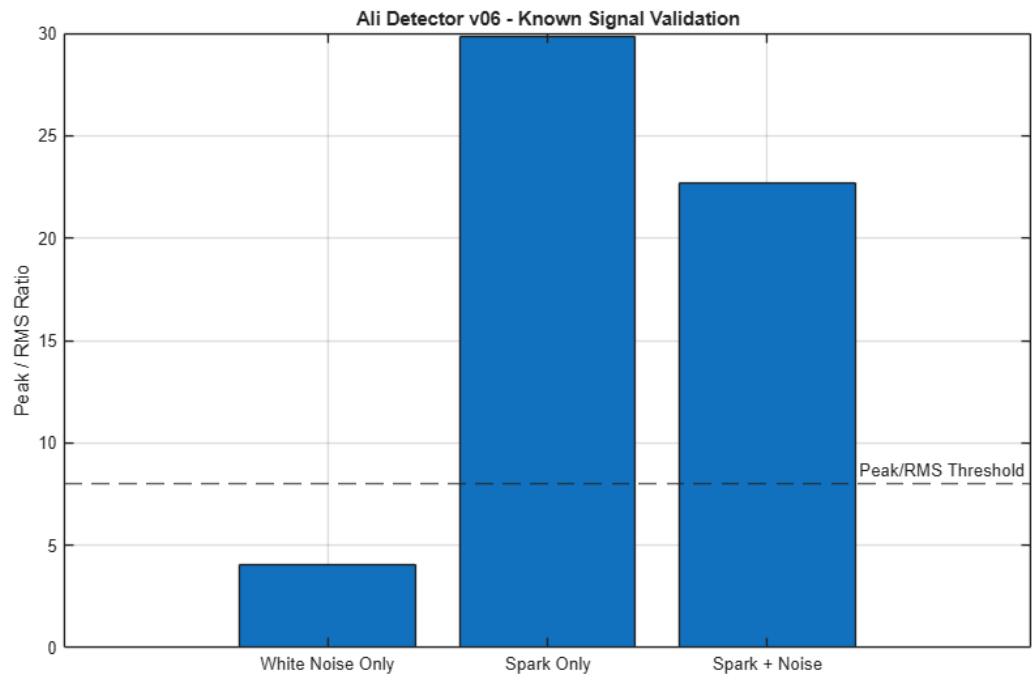
```

set(gca, ...
    'XTick', 1:numFiles, ...
    'XTickLabel', fileLabels);

ylabel('Peak / RMS Ratio');
title('Ali Detector v06 - Known Signal Validation');
grid on;

yline(8, '--', 'Peak/RMS Threshold');

```



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